CAS2105 Homework 6

Sentiment Analysis of Running Belt Reviews: Comparing Keyword-based Baseline and AI Pipeline

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1 Task Definition

- **Task description:** The objective of this project is to perform binary sentiment classification on customer reviews for running belts. Specifically, the system takes a Korean review text as input and classifies it into two categories: Positive (1) or Negative (0).
- **Motivation:** I recently purchased a running belt from an e-commerce platform (Naver Plus Store) and spent a significant amount of time manually reading through reviews to decide whether to buy it. This personal experience highlighted how crucial reviews are for decision-making for buyers. I realized that developing an AI pipeline to automatically analyze these reviews would be an interesting application of AI.
- **Input / Output:** The input is a Korean text string representing a user review. A balanced dataset of 100 examples (50 positive and 50 negative) was sampled from approximately 5,880 raw reviews collected from an e-commerce platform for a running belt product. The output is a binary label, where 1 indicates Positive and 0 indicates Negative. This prediction is compared against the ground truth label derived from the user's original star rating.
- **Success criteria:** The system is considered "good" if it achieves high classification accuracy when compared to the actual user ratings. A key success criterion is whether the improved AI pipeline can significantly outperform the accuracy of the simple Naïve Baseline.

2 Methods

2.1 Naïve Baseline

- **Method description:** I implemented a keyword-based text classification method. I defined a specific list of negative keywords associated with running belts (e.g., "uncomfortable," "broken," "shake," and "worst"). The model scans the input text; if any of these negative keywords are found, it predicts Negative (0). If none are found, it defaults to Positive (1).
- **Why naïve:** This approach is considered naïve because it relies on simple string matching without understanding the semantic context or sentence structure. It treats the text as a "bag of words" and ignores how words interact with each other.
- **Likely failure modes:** The model is likely to fail in handling negation. For example, a review stating "안 흔들려요" (It does not shake) is a positive review for a running belt. However, the baseline will detect the keyword "shake" (흔들) and incorrectly classify it as Negative because it cannot understand that the word "not" (안) reverses the meaning.

2.2 AI Pipeline

- **Models used:** I utilized a pre-trained transformer model from Hugging Face: matthewburke/korean_sentiment. This model is based on the multilingual BERT (mBERT) architecture and has been fine-tuned on the Naver Sentiment Movie Corpus (NSMC), which consists of 150,000 Korean reviews.
- **Pipeline stages:**
  1. **Preprocessing:** The raw Korean text is converted into tokens (numbers) that the model can understand. This step is handled automatically by the Hugging Face pipeline tokenizer.
  2. **Representation & Inference:** The tokenized input is passed through the pre-trained Transformer layers to generate contextual embeddings. The model then outputs logits for two classes (LABEL_0 and LABEL_1).
  3. **Decision Component:** The system interprets the model's output. It maps LABEL_1 to Positive (1) and LABEL_0 to Negative (0) to match our dataset's format.
- **Design choices and justification:** I chose to use a pre-trained model instead of training a model from scratch for the following reasons.
  1. **Small Dataset:** Since my collected dataset contains only 100 examples, it is too small to train a deep neural network effectively.
  2. **Domain Similarity:** The chosen model was trained on movie reviews. While my task is about "running belts," the emotional vocabulary used to express satisfaction (e.g., "good," "recommend") or dissatisfaction (e.g., "bad," "worst") is highly similar across both domains. Therefore, the pre-trained knowledge transfers well to this new task without additional fine-tuning.

3 Experiments

3.1 Datasets

- **Source:** The raw dataset was collected from the *Fit Flip Belt Running Belt* product page of the *Super Mommy* store on Naver Plus Store[1], utilizing the *Smart Review* Chrome extension.
- **Total examples:** I curated a balanced dataset of 100 examples (50 Positive, 50 Negative) sampled from the raw dataset.
- **Train/Test split:** Since this is a mini-project focusing on inference capability, I used the entire 100-sample dataset as a test set to compare the baseline and the AI pipeline directly.
- **Preprocessing steps:** The text data was cleaned to remove encoding errors, but no specific normalization (like stemming) was applied, as BERT-based models handle raw text effectively.

3.2 Metrics

- **Accuracy:** Since the dataset is perfectly balanced (50:50), accuracy is a fair and sufficient measure to evaluate how many reviews were correctly classified against the user's actual star ratings.

3.3 Results

- **Quantitative Evaluation:** The experimental results demonstrate a significant performance gap between the two approaches. As shown in the table below, the AI Pipeline outperformed

[1] Website URL: `https://vo.la/aVKcBRl`

the Naïve Baseline in terms of accuracy.

Method	Accuracy
Baseline	0.54
AI Pipeline	0.86

Table 1: Performance Comparison between Naïve Baseline and AI Pipeline

- **Qualitative Analysis:** The AI Pipeline successfully classified complex instances where the baseline failed. Notable examples of these success cases are presented below.

  1. **Indirect Expression (typos included)**
     Review: "개인 체격에 맞게 주문하여야 할 꺼 갇습니다"
     (Translation: I think you need to order according to your body size.)
     → **Prediction: Correct (0, Negative)**
     Analysis: The model correctly interpreted the suggestion as a negative sizing issue despite the typos.

  2. **Short & Direct Negative**
     Review: "불친절해요"
     (Translation: [The seller is] unfriendly.)
     → **Prediction: Correct (0, Negative)**
     Analysis: The model accurately detected the strong negative sentiment associated with the specific keyword "unfriendly," demonstrating its robustness even with extremely short inputs.

  3. **Long & Comparative Positive**
     Review: "다른 제품이랑 큰 차이 없네요... (중략) ... 잘 흘러내리지도 않고 물통도 넣어서 다닐 수 있습니다"
     (Translation: There isn't much difference compared to other products... It fits well and holds a water bottle.)
     → **Prediction: Correct (1, Positive)**
     Analysis: The model handled long context and identified the praise amidst comparative sentences.

4 Reflection and Limitations

- **Successes and Failures:** The experiment confirmed that the AI pipeline handles semantic context significantly better than the keyword-based baseline, particularly in processing negations.

  1. **Success Case:** For the review "하나도 불편하지 않고 딱 고정되서 너무 좋네요" (It is not uncomfortable at all and stays fixed), the Baseline incorrectly predicted Negative because it detected the keyword "uncomfortable" (불편). However, the AI model correctly predicted Positive by understanding the context of "not uncomfortable."

  2. **Failure Case:** The AI model struggled with specific functional descriptions lacking strong emotional keywords. For example, in the review "자꾸 핸드폰이 빠져요" (The phone keeps falling out), the AI predicted Positive. This is likely because the sentence describes a factual defect without using typical negative sentiment words like "worst" or "bad," suggesting the model relies heavily on emotional tone rather than functional context.

- **Metric Suitability:** Accuracy was selected as the primary metric. Given the balanced dataset (50 Positive, 50 Negative), accuracy provided a fair and effective performance comparison between the rigid rule-based approach and the probabilistic AI approach.

- **Limitations:** A key limitation is the scale of the evaluation set (100 examples). While restricted for thorough manual verification, this small size may not fully represent the diversity of the entire dataset.
- **Future Work:** Future work will involve applying this pipeline to the full dataset to analyze macro-trends. I also plan to fine-tune the model on sports equipment reviews to better capture domain-specific complaints (e.g., "bouncing," "zipper failure") that general sentiment models might miss.